

DAY-1

Tasks

1.

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the multimeter helps
Computer won't turn on	Checks the PSU output voltage to see whether the power supply is providing the correct voltage.
Laptop battery isn't charging	Measures the battery/charger voltage to determine whether power is reaching the battery correctly.
Smartphone won't charge	Checks voltage/current at the charging port or charging circuit to identify a faulty charger, port, or power circuit.

2.

Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

A POST card is inserted into the motherboard to monitor the Power-On Self-Test process. During startup, its hexadecimal display changes as the BIOS checks different hardware components. In the demonstration, the card displays different POST codes during the boot process, and the final code indicates that the startup tests have completed successfully. POST codes can be used to identify where startup stops, helping diagnose hardware such as the CPU, RAM, graphics system, or motherboard.

A useful reference/demo is the ITFreeTraining POST-card demonstration, which specifically shows the code changing during startup and explains how the last displayed code can indicate the hardware test where a failure occurred.

3.

Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
SIDDIHI				
LENOVO LNVN8161216				
Temperatures				
\TZ.TZ00	28.0 °C	28.0 °C	28.0 °C	
Intel PCH	45.0 °C	45.0 °C	46.0 °C	
Utilization				
Physical Memory Load	75.0 %	75.0 %	75.0 %	
Available Physical Memory	3.8 GB	3.7 GB	3.8 GB	
Virtual Memory Load	65.1 %	65.1 %	65.4 %	
Available Virtual Memory	7.7 GB	7.7 GB	7.7 GB	
Intel Core Ultra 5 125H				
Voltages				
VID (Max)	0.795 V	0.728 V	0.868 V	
VR Cores VID	0.690 V	0.615 V	0.790 V	
VR GT VID	0.000 V	0.000 V	0.515 V	
VR SA VID	0.500 V	0.485 V	0.865 V	
LLC/Ring	0.363 V	0.350 V	0.363 V	
Temperatures				
P-Cores (Max)	50.0 °C	50.0 °C	57.0 °C	
E-Cores (Max)	51.0 °C	50.0 °C	54.0 °C	
LP-Cores (Max)	53.0 °C	53.0 °C	57.0 °C	
Package	69.0 °C	54.0 °C	69.0 °C	
VR Limit	125.0 °C	125.0 °C	125.0 °C	
VR Cores	50.0 °C	49.0 °C	52.0 °C	
VR GT	50.0 °C	49.0 °C	52.0 °C	
VR SA	50.0 °C	49.0 °C	52.0 °C	
Powers				
Package	15.71 W	3.53 W	15.71 W	
IA Cores	10.95 W	0.52 W	10.95 W	
GT	0.01 W	0.00 W	0.22 W	
Power Max (PL1)	40.00 W	40.00 W	40.00 W	
Short Power Max (PL2)	100.00 W	100.00 W	100.00 W	
Max Peak Power (PL4)	120.00 W	120.00 W	120.00 W	
Currents				
IccMax	110.00 A	110.00 A	110.00 A	
VR Cores Output	5.61 A	1.73 A	8.63 A	
VR GT Output	0.00 A	0.00 A	5.33 A	
VR SA Output	1.10 A	1.10 A	3.98 A	
Clocks				
P-Cores (Max)	1995.2 MHz	1795.5 MHz	2494.2 MHz	
E-Cores (Max)	1995.2 MHz	1396.5 MHz	2893.0 MHz	
LP-Cores (Max)	2495.2 MHz	2495.2 MHz	2495.2 MHz	

4.

Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Electrical voltage, resistance, and continuity	Faulty power supply or charging circuit
POST Card	Motherboard startup/POST process using diagnostic codes	RAM, CPU, BIOS, or motherboard startup failure
Software Diagnostics	Hardware temperatures, voltages, storage health, and system status	Overheating CPU or failing hard drive/SSD